San Diego Airbnb
STRO Ordinance Analysis

Did the May 2023 short-term rental cap change prices differently in high-Airbnb-density areas?

Difference-in-Differences · OLS · EDA

MATH 189 · Group 37
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Final Project Presentation

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Background: why this matters

Short-term rentals are often blamed for removing housing supply in tourist-heavy neighborhoods.

4th

San Diego ranked as one of the least affordable U.S. cities in the report we cited.

1%

May 2023 STRO ordinance capped whole-home short-term rentals at 1% of city housing stock.

~50%

Policy was described as removing nearly half of short-term rentals citywide.

Why it is interesting: if many STRs were converted back into housing, high-Airbnb-density areas might see different price or rent changes than low-density areas.

Policy shock

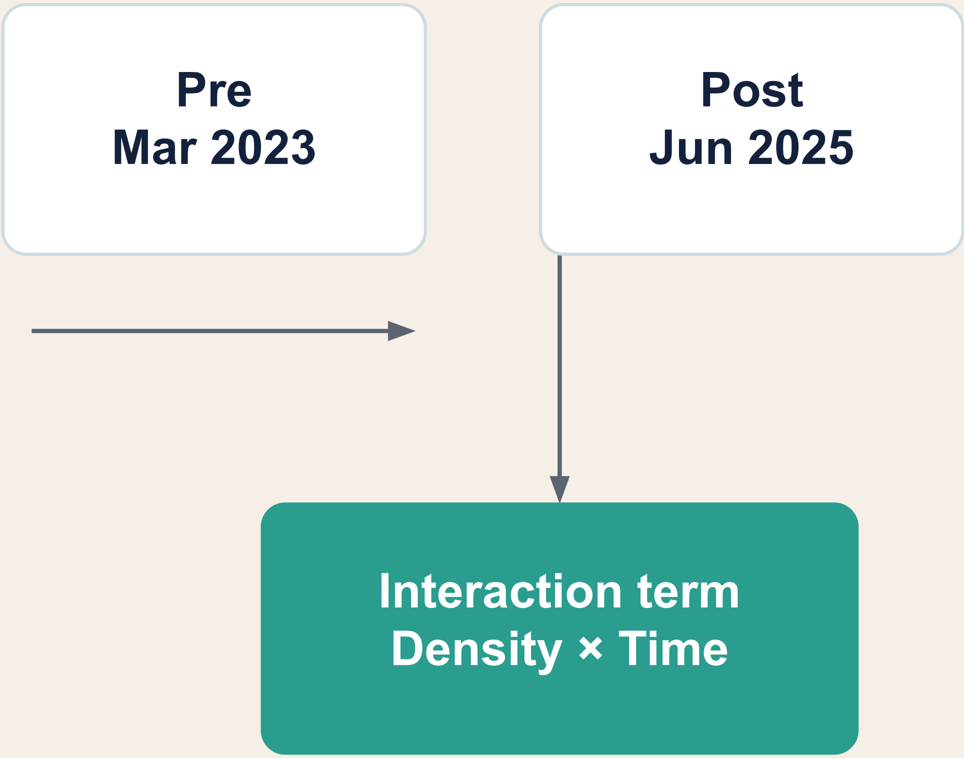
A clear before/after date makes this a good setting for comparing changes across groups.

May 1, 2023

Research question & hypothesis

We compare high-density and low-density Airbnb areas before and after the ordinance.

Did the STRO ordinance affect prices differently in high-Airbnb-density areas compared with low-density areas?



Null: no differential effect between high- and low-density areas
Alternative: price/rent changes differ by Airbnb density

Data sources & cleaning

Two datasets were combined to study the short-term and long-term housing markets.

Inside Airbnb
March 2023 vs. June 2025
Fields: price, neighborhood, latitude, longitude, room type

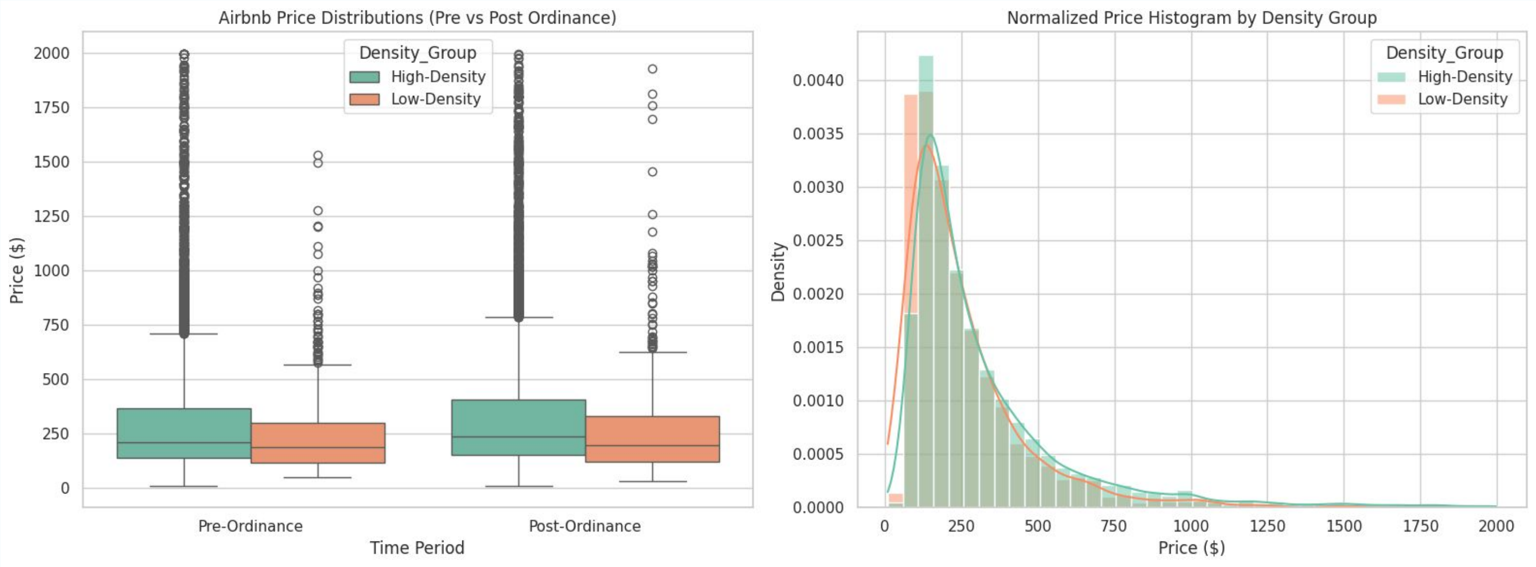
Zillow ZORI
Jan 2023 vs. Jan 2025
ZIP-code rent index for San Diego

Cleaning choices
Kept “Entire home/apt” only; trimmed top 1% luxury outliers above $1,999/night; log-transformed prices.

Final analysis datasets
- Airbnb log-price model: 20,526 listings after filtering
- ZORI rent model: San Diego ZIP-code rent observations

EDA: price distributions are skewed

We first checked how Airbnb prices differ by density group and time period.



Main EDA takeaways
• Prices are strongly right-skewed
• A small number of luxury listings create heavy tails
• High-density areas contain most listings in the sample

19,373

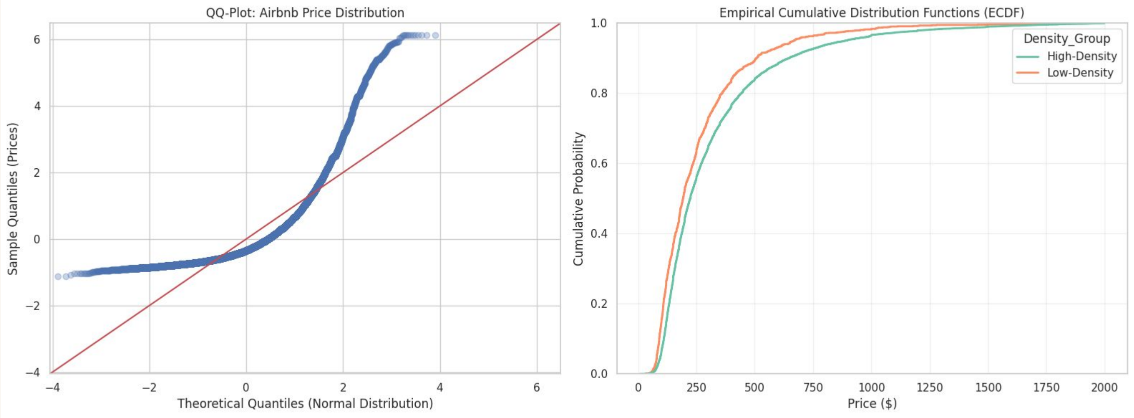
High-density listings

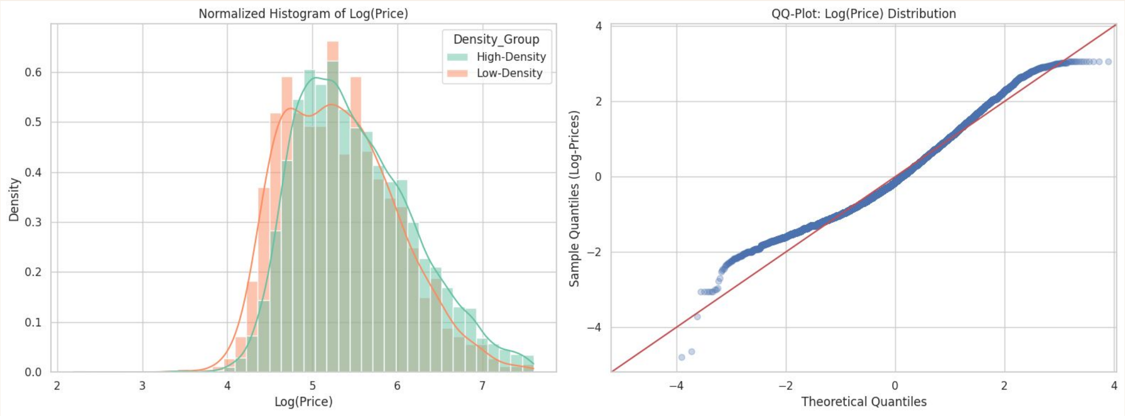
1,357

Low-density listings

Why we used log(price)

The raw price data had heavy tails, so we used a log transformation before OLS.





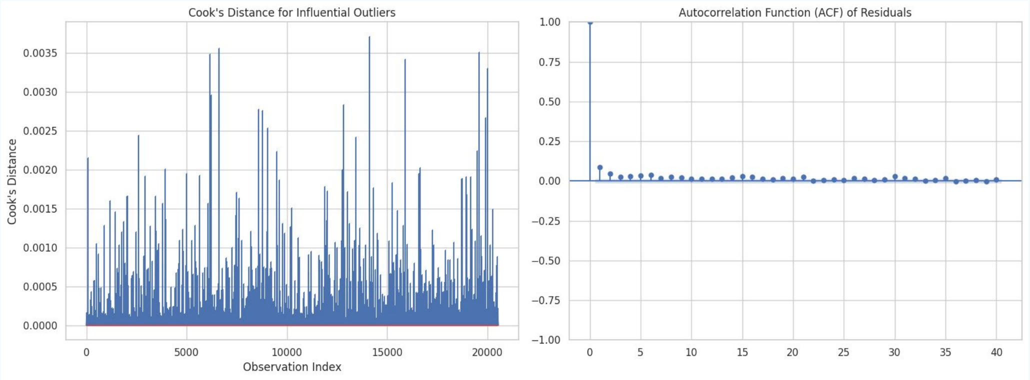
Interpretation: after using log(price), regression coefficients can be read approximately as percent changes.

Statistical method

Our main model used an interaction term to test the differential policy effect.

Model idea

log(price or rent) = density group + time period + density × time

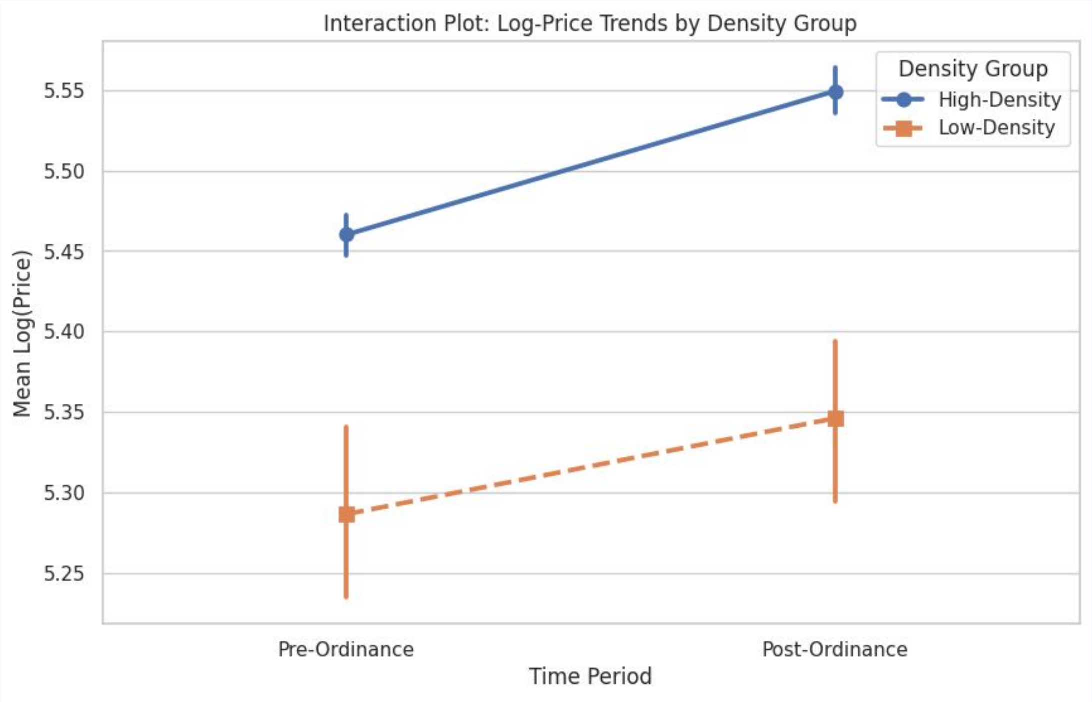


OLS regression for Airbnb nightly prices
OLS regression for Zillow long-term rent
Cook's distance and ACF plots for diagnostics
KS test used to compare distributions when normality was questionable

Key term: if the interaction p-value is small, the policy effect differs between high- and low-density areas.

Results: Airbnb nightly prices

The remaining Airbnbs did not show a statistically significant differential price change.



0.0297

Interaction coefficient for Airbnb log-price

0.444

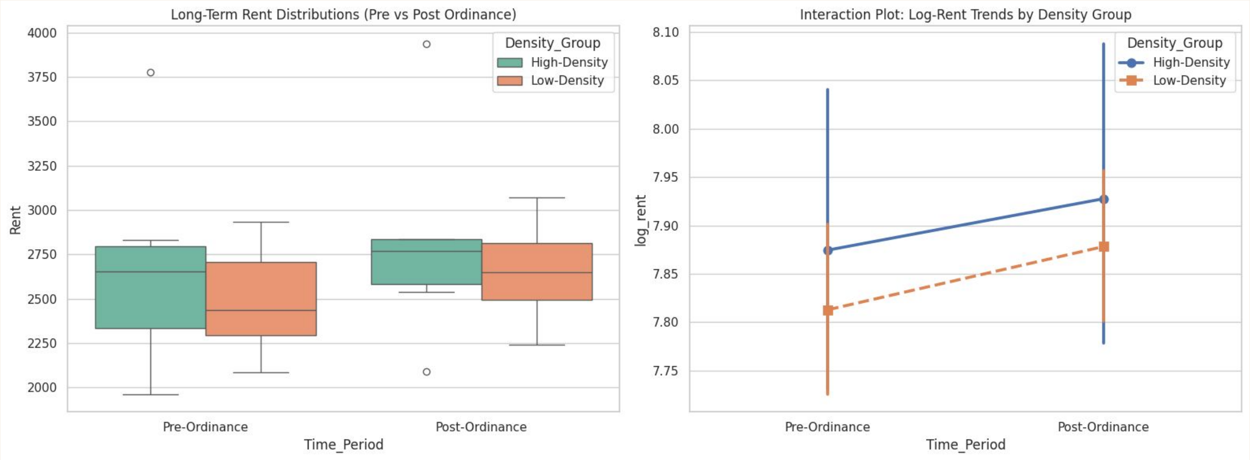
Interaction p-value

Interpretation: high-density areas had about a 3% larger estimated change, but the p-value is much higher than 0.05, so we fail to reject the null hypothesis.

KS test: high- and low-density price distributions are different overall ($p = 1.82e-15$), but this does not prove a policy effect by itself.

Results: long-term rents (ZORI)

We also tested whether long-term rents changed differently in high-Airbnb-density ZIP codes.



-0.0122

Interaction coefficient for log-rent

0.930

Interaction p-value

Interpretation: the estimated rent change in high-density ZIP codes was about 1.2% lower, but the p-value is extremely high, so the result is not statistically significant.

Conclusion for ZORI: no clear evidence that the ordinance produced stronger long-term rent relief in high-Airbnb-density ZIP codes.

Conclusion & limitations

The data did not show a clear differential price effect from the ordinance.

Main conclusion
We fail to reject the null hypothesis for both Airbnb prices and long-term rents. The ordinance may have reduced STR inventory, but our data does not show measurable rent relief in high-density areas.

Limitations
- Only one pre and one post Airbnb snapshot
- ZORI is ZIP-code level, less precise than neighborhood or listing coordinates
- Two-year gap includes inflation, interest rates, and tourism changes

What we would do next
Use more monthly snapshots, smaller geographic units, and stronger controls for broader market trends.

Sources: Inside Airbnb, Zillow ZORI, Chapman University, Garcia-López et al. (2020), Duso et al. (2020)